

Smart Farm Asset Inspector

1. Chosen use case

Asset inspection (Challenge Rules §3). The assets inspected are crop rows on a working farm; the inspecting platform is a Unitree Go2 quadruped running the Eagle ROS 2 navigation stack.

2. Who this is for

Primary recipient: mid-size specialist crop growers — 20 to 200 hectares of high-value row crops such as tomatoes, lettuce and brassicas, in open field or under tunnel. This segment is large enough to carry a per-hectare crop loss that justifies equipment, and small enough that it has no in-house agronomy team.

Secondary: agricultural cooperatives and contract agronomy services, who would operate one robot across several member holdings.

3. The operational problem

Crop disease and stress are found by walking the rows and looking. That scouting is done by the grower or a visiting agronomist, and it has three structural weaknesses:

It is sampled, not systematic. A scout walks a fraction of the rows and extrapolates. Infection that starts in an unwalked row is found late.

It is irregular. Scouting competes with every other task in the working day and is the first thing dropped in a busy week — which tends to be exactly the week that matters.

It is not recorded. Findings live in memory or a notebook. There is rarely a georeferenced image trail showing what a given row looked like a fortnight ago, so progression cannot be judged and treatment cannot be audited.

The consequence is late detection. For most foliar diseases the cost of intervention rises steeply with delay, because the treatable window closes and the affected area grows.

Supporting evidence. This is a well-documented pattern in the precision agriculture literature, and the robotics response — autonomous scouting platforms in row crops — is an established and growing product category. We have not run a customer study, and this brief does not claim one; the need described here should be validated with two or three growers before any commercial commitment. Stating that plainly is more useful than citing figures we cannot stand behind.

4. What the solution does, from the user's point of view

The grower types or speaks a request in ordinary language:

"Check the north field for powdery mildew on the tomatoes."

Within about four seconds the system has interpreted the request, planned a route over the mapped rows of that field, checked the plan against the robot's current state, and started the robot walking. The

robot photographs the crop at fixed intervals, avoids obstacles and people on its own, and returns to its dock. The imagery is then classified for plant health and the grower receives a list of anomalies.

Three things the grower never does: open a mapping tool, place a waypoint, or learn any syntax.

When the request cannot be executed, the system says so instead of guessing. Asked to inspect a field that does not exist, it replies:

“‘West orchard’ is not a field on Demo Farm North. Did you mean north_field, central_field, or south_field?”

Asked to inspect the entire farm on the available battery, the deterministic validator rejects the plan before the robot moves and explains that the mission needs splitting.

5. Why it is scalable

Scalability is credible only if the split between reusable and rebuilt parts is explicit. Ours:

Reused unchanged on a second customer or second site — roughly 95% of the code

Component	Why it transfers
LLM interpretation layer	Prompt is assembled at runtime from the site’s own config; no per-site prompt engineering
Deterministic validator	Rules are expressed against the site model, not against a specific farm
Task generator and mission compiler	Emit the same <code>MissionData</code> any Eagle-stack robot consumes
ROS 2 bridge and safety integration	Standard interfaces
Vision detector	Colour-based, no per-site training
Test suite (39 tests)	Site-independent

Rebuilt per site — roughly 5%, and none of it is engineering

Item	Effort
<code>field_map.yaml</code> — fields, rows, aliases, geofence, no-go zones	Half a day, from an existing farm map
Occupancy map for the robot	One SLAM run, standard commissioning
Detector threshold calibration	Half a day against local lighting and crop

The reusable part contains all the engineering; the rebuilt part is configuration a commissioning technician can do. That ratio is what makes the second deployment cheap, and it is the reason to build the interpretation layer generically rather than hard-coding one farm.

The vocabulary transfers too. Nothing in the parser knows what a tomato is. Crops, inspection types and field names come from config, so moving to vineyards or orchards means editing YAML, not code. The same architecture was prototyped against a vehicle depot during development, and the planner, validator, compiler and ROS bridge carried over unchanged.

6. What is not yet solved

- Detection identifies chlorosis — leaf yellowing — not a named

disease. Distinguishing powdery mildew from drought stress needs either a trained classifier or a multispectral sensor.

- Per-plant false positive and negative rates are not measured; only aggregate agreement with simulated ground truth.
- Simulation only. No hardware trial.
- The energy model is uncalibrated.